

An investigation of changes in taste perception by varying polygon resolution of foods in virtual environments

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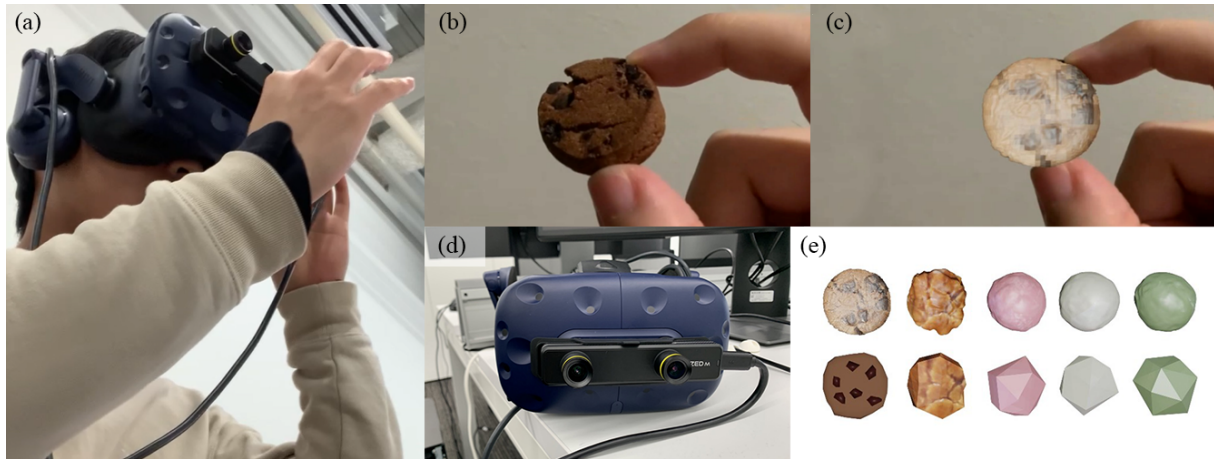


Figure 1: Diagram of our experiment. (a): Experiment participants are wearing a VR device, eat foods, and answer a questionnaire. (b): Subjects eat a cookie without image processing and check the taste and texture. (c): Eating cookies with projected a CG image. (d): An external camera was attached to the HMD for image recognition. (e): CG model used in the experiment. The upper/lower row is a high/low resolution model, respectively. From left to right: cookies, Okaki, and Dango (three colors).

ABSTRACT

In recent years, metaverse has received considerable attention. We believe that as this technology develops, humanity can dine in a virtual space while maintaining a sense of immersion. Therefore, we investigated whether the taste of food is influenced by the number of polygons of CG models using VR/AR technology. We created CG models and overlaid the image onto the actual food via HMD. Then the subjects consumed the food which CG image overlaid and answered a questionnaire. Results showed that the higher the number of polygons, the less hardness was perceived and the toon-like model was more likely to affect the taste.

CCS CONCEPTS

• Computing methodologies → Mixed / augmented reality.

KEYWORDS

VR; AR; Taste perception

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1 INTRODUCTION

Recently, as more and more people spend long hours in virtual spaces, research on eating in virtual spaces is gaining momentum. In particular, research using AR technology to change the appearance of food products to other foods and investigate how the changes affect taste has garnered a lot of attention [Nakano et al. 2019].

We focused on the visual and memory processing of CG models related to eating in virtual space, with particular attention to the number of polygons in CG models. The appearance of a CG model is definitely determined by the number of polygons and texture resolution. While there have been previous studies on changes in taste due to color variation [Huang et al. 2019], there have been no studies focusing on changes in the number of polygons. We investigated what kind of CG models would allow humans to eat without discomfort, and whether there would be any problems when eating in a virtual space.

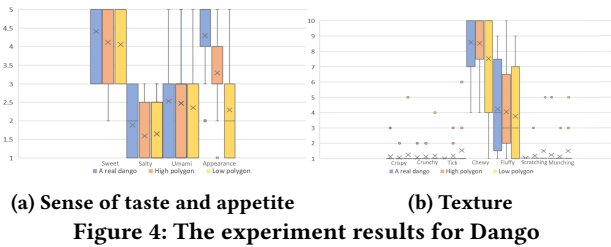
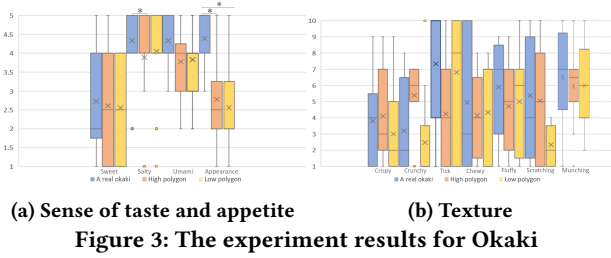
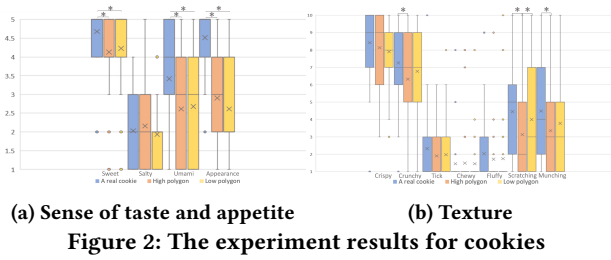
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2 MATERIALS & METHODS

The foods used in the experiment were commercially available snacks, as they differed in texture, hardness, and taste and should be familiar to the subjects. In addition, the snacks could be eaten in one bite so that shape changes after eating would not have to be taken into consideration. Cookies, Okaki (soy sauce-flavored), and Dango were selected and we created CG models at high and low resolution (reduced to less than 1% of the high resolution model), respectively. For the cookies, in addition to the high-polygon and low-polygon models, we also compared the low-polygon model with the texture changed to a toon style (created by reducing the resolution from 500x500 to 30x30). Figure 1 (b) and (c) show images of cookies with and without CG models, and Figure 1 (e) shows CG models created to correspond the each food used in the experiment.

We employ VIVE PRO (HMD) and ZED Mini (External camera) (Figure 1 (d)). VR devices and high quality cameras are expected to improve the accuracy of object recognition and the quality of experiments. The food was recognized through the ZED Mini, and visual information was overwritten by sending the CG model over the real food on the HMD. YOLACT++ was used for the recognition.

At first, the subjects looked at the snack without wearing the HMD and then to eat the target food once to memorize the sensation. Afterward, the subjects wore the HMD, picked up the food with their index fingers and thumbs, held it in front of the camera (Figure 1 (a)). Our system overlaid the CG image on recognized food. It is performed twice, using the image of the high-polygon/low-polygon model. A total of 66 experiments (cookies: 31, Okaki: 18, Dango:

17) were conducted, with questionnaires from a wide range of ages, from under 10 to 60s. The questionnaire is used to evaluate the taste ("Sweet," "Salty" and "Umami"), and whether the appearance induced appetite in a 5-point scale, and whether the onomatopoeia ("Crispy," "Scratching," "Tick," "Chewy," "Fluffy," "Crunchy" and "Munching") was applicable to the texture in a 10-point scale. The results obtained were subjected to a Wilcoxon signed-rank test. We compare the results of eating without the CG model and eating while looking at the high-polygon/low-polygon CG model, respectively.

3 RESULTS

As shown in Figure 2, significant differences were found between the real cookies and the high polygon model in the items of "Appearance," "Sweet," "Umami," "Scratching," "Crunchy," and "Munching" for the cookies (* : $p < 0.05$). Significant differences were found between the real cookies and the low-polygon model for "Appearance," "Sweet," and "Umami" taste. In addition, significant differences were also observed in "Scratching", depending on the polygon. The toon-like model looks completely different from the low-polygon model, giving it a cheap impression. Therefore, the difference between the toon-like model and the model without any overwriting was noticeable. It was found to not feel less hard when the number of polygons was high, as "Scratching," "Crunchy," and "Munching" are onomatopoeic expressions of hardness.

There were significant differences in "Appearance" and "Salty" items between real and high polygons for Okaki, and also a significant difference in "Salty" item between real and low polygons (Figure 3). For Dango, although no significant differences were found for all items, differences were identified for chewy and fluffy items (Figure 4). These results confirm that overriding the CG model has the effect of blunting the originally existing taste and texture.

4 CONCLUSIONS

We focused on whether taste changes due to the resolution of CG model and evaluated whether there was a change in taste when the CG model was displayed on a real food via an HMD in eating. It was found that the 3D model made the food less appetizing than when viewed directly, and reduced the taste sensation best experienced when eating the real food. We found that a smaller number of polygons made the food feel harder, and that the lower the resolution of the texture, the greater the negative effect on taste.

Future tasks are to gradually reduce the number of polygons and investigate compromises, and to experiment not only with the number of polygons but also with different rendering methods.

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REFERENCES

- Fuxing Huang, Jianping Huang, and Xiaogang Wan. 2019. Influence of virtual color on taste: Multisensory integration between virtual and real worlds. *Computers in Human Behavior* 95 (2019), 168–174. <https://doi.org/10.1016/J.CHB.2019.01.027>
- Kizashi Nakano, Daichi Horita, Nobuchika Sakata, Kiyoshi Kiyokawa, Keiji Yanai, and Takuji Narumi. 2019. Enchanting Your Noodles: GAN-based Real-time Food-to-Food Translation and Its Impact on Vision-induced Gustatory Manipulation. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 1096–1097. <https://doi.org/10.1109/VR.2019.8798336>